

Experiment 11: Credit Score Classification**Aim**

To implement a machine learning model for the task of **Credit Score Classification**.

Algorithm

- Step 1:** Import the required libraries.
- Step 2:** Load the credit score dataset.
- Step 3:** Handle missing values in the dataset.
- Step 4:** Encode categorical features into numerical values.
- Step 5:** Separate the input features and target variable.
- Step 6:** Split the dataset into training and testing data.
- Step 7:** Train a suitable classification model.
- Step 8:** Predict the credit score class for the test data.
- Step 9:** Evaluate the model using accuracy.

Program

```
# Experiment 11 - Credit Score Classification

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

data = pd.read_csv("credit_score.csv")

data = data.dropna()

encoder = LabelEncoder()

for column in data.select_dtypes(include="object").columns:
    data[column] = encoder.fit_transform(data[column])

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Output

```
... Predictions: [0 0 1 1 0 1 0 0 0 1 1 1 0 1 0 0 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0
 1 1 0 0 1 0 0 1 0 1 0 1 0 0 1 0 0 1 0 0 0 0 0 0 0 0 1 0 0 0 0 0 1 1 1 1 0 0
 0 1 1 1 0 0 0 0 0 1 0 1 1 0 0 0 0 1 0 1 1 0 0 0 1 1 0 0 0 1 1 0 0 0 1 0 1
 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 1 1 0 0 1 1 0 0 1 0 0 1 0
 1 1 0 0 1 0 1 1 1 0 1 0 0 1 1 0 0 1 0 0 0 1 0 0 1 0 0 0 0 0 1 0 1 0 0 0 1
 1 0 0 0 1 1 1 1 0 1 0 1 1 1 0]
Accuracy: 0.61
```

Result

Thus, the Credit Score Classification model was successfully implemented, and the credit score categories were predicted using the trained Decision Tree classifier.